2. The hypothalamus
   (A) synthesizes all of the hormones produced by the pituitary gland.
   (B) influences the function of only one lobe of the pituitary gland.
   (C) produces only inhibitory hormones.
   (D) regulates both reproduction and body temperature.

3. Growth factors are local regulators that
   (A) are produced by the anterior pituitary.
   (B) are modified fatty acids that stimulate bone and cartilage growth.
   (C) are found on the surface of cancer cells and stimulate abnormal cell division.
   (D) bind to cell-surface receptors and stimulate growth and development of target cells.

4. Which hormone is *correctly* paired with its action?
   (A) oxytocin—stimulates uterine contractions during childbirth
   (B) thyroxine—inhibits metabolic processes
   (C) ACTH—inhibits the release of glucocorticoids by the adrenal cortex
   (D) melatonin—raises blood calcium level

Levels 3-4: Applying/Analyzing

5. What do steroid and peptide hormones typically have in common?
   (A) their solubility in cell membranes
   (B) their requirement for travel through the bloodstream
   (C) the location of their receptors
   (D) their reliance on signal transduction in the cell

6. Which of the following is the most likely explanation for hypothyroidism in a patient whose iodine level is normal?
   (A) greater production of T_3 than of T_4
   (B) hyposecretion of TSH
   (C) hypersecretion of MSH
   (D) a decrease in the thyroid secretion of calcitonin

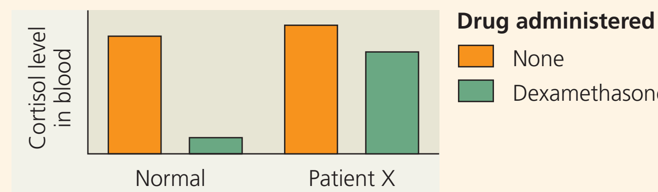
7. The relationship between the insect hormones ecdysteroid and PTTH is an example of
   (A) an interaction of the endocrine and nervous systems.
   (B) homeostasis achieved by positive feedback.
   (C) homeostasis maintained by antagonistic hormones.
   (D) competitive inhibition of a hormone receptor.

8. **DRAW IT** In mammals, milk production by mammary glands is controlled by prolactin and prolactin-releasing hormone. Draw a simple sketch of this pathway, including glands, tissues, hormones, routes for hormone movement, and effects.

Levels 5-6: Evaluating/Creating

9. **EVOLUTION CONNECTION** The intracellular receptors used by all the steroid and thyroid hormones are similar enough in structure that they are all considered members of one "superfamily" of proteins. Propose a hypothesis for how the genes encoding these receptors may have evolved. (Hint: See Figure 21.13.) Explain how you could test your hypothesis using DNA sequence data.

10. **SCIENTIFIC INQUIRY • INTERPRET THE DATA** A chronically high level of glucocorticoids can result in obesity, muscle weakness, and depression, a combination of symptoms called Cushing's syndrome. Excessive activity of either the pituitary or the adrenal gland can be the cause. To determine which gland has abnormal activity in a particular patient, doctors use the drug dexamethasone, a synthetic glucocorticoid that blocks ACTH release. Based on the graph, identify which gland is affected in patient X.



11. **WRITE ABOUT A THEME: INTERACTIONS** In a short essay (100–150 words), discuss the role of hormones in an animal's responses to changes in its environment. Use specific examples.

12. **SYNTHESIZE YOUR KNOWLEDGE**

The frog on the left was injected with MSH, causing a change in skin color within minutes due to a rapid redistribution of pigment granules in specialized skin cells. Using what you know about neuroendocrine signaling, explain how a frog could use MSH to match its skin coloration to that of its surroundings.

For selected answers, see Appendix A.